

Convex Optimization Exercises - Sheet 1

Exercise 0. Install CVXPY. If you have Python installed on your computer, then you may follow the instructions at <https://www.cvxpy.org/install/index.html> (e.g. by running `pip install cvxpy`).

If you prefer to compile Python without having it installed locally, then you may use <https://jupyter.org/>. Go to “Try” → “JupyterLab”, and there you may create a new “Notebook”. The website is free and requires no registration.

Exercise 1. both by hand and by using CVXPY — compare the results. Observe how we introduce variables, objective functions, constraints, and optimization problems in CVXPY.

$$\begin{aligned} & \text{minimize} && f(x, y) = x + y \\ & \text{subject to} && 2x + y \geq 2 \\ & && x + 2y \geq 2 \end{aligned}$$

Exercise 2. Try for yourself to solve the following problem using CVXPY.

$$\begin{aligned} & \text{minimize} && f(x, y, z) = 4x + 5y + 3z \\ & \text{subject to} && x^2 + 2y^2 + z^2 \leq 4, \end{aligned}$$

(*) Try computing the exact solution by hand and compare the results.

Exercise 3. Consider the problem of the ‘optimal diet’, from the first lecture:

$$\begin{aligned} & \text{minimize} && f(x) = c^T x \\ & \text{subject to} && Ax \succeq b \\ & && x \succeq 0, \end{aligned}$$

where

$$A = \begin{pmatrix} 35 & 0.5 & 0.5 \\ 60 & 300 & 10 \\ 30 & 20 & 10 \end{pmatrix}, \quad b = \begin{pmatrix} 0.5 \\ 15 \\ 4 \end{pmatrix}, \quad c = \begin{pmatrix} 0.75 \\ 0.5 \\ 0.15 \end{pmatrix}.$$

Solve this problem using CVXPY, specifically getting familiar with how vectors/matrices are encoded in NumPy / CVXPY.

Exercise 4. Come up with examples of a linear program (or other convex problem), such that (i) it has no solution, (ii) it has an optimal solution that is not unique, (iii) it is unbounded. What does CVXPY output as optimal value, optimal solution, and ‘status’ of these problems?

Have a look at the documentation of CVXPY <https://www.cvxpy.org/tutorial/index.html>. In particular, check which constraint and objective functions can be encoded (<https://www.cvxpy.org/tutorial/functions/index.html>).

Remark

The main advantages of CVXPY are that:

- it is free;
- it is relatively intuitive to use (using notation that is close to mathematical notation);
- it can handle small and medium sized problems relatively quickly;
- it was co-developed by Boyd, and therefore aligns well with the material of this course.

However, depending on the application, other software might be more helpful. There is a very big variety of optimization software, both for free and commercial use — see e.g. https://en.wikipedia.org/wiki/List_of_optimization_software. Another prominent optimization package for Python is SciPy, which includes more general-purpose tools that can also be used to solve some non-convex optimization problems; this comes however at the cost of more manual calibrations and low-level understanding of algorithms.

Also mathematics software that you might already be familiar with (MATLAB, Mathematica, Maple) has built-in optimization tools — we are not going to discuss them in this course — however, feel invited to experiment with them and share what you think are their advantages/disadvantages.